

Exploiting Hybrid Metasurfaces for enhanced Linear and Nonlinear Chiral Sensing Applications

Guillermo Serrera, Javier González-Colsa and Pablo Albella*

Department of Applied Physics, University of Cantabria, Avda. Los Castros, s/n, 39005 Santander, Spain

[*pablo.albella@unican.es](mailto:pablo.albella@unican.es)

The concept of chirality, denoting the absence of specular symmetry in any plane of an object and the consequent existence of two mirror images (enantiomers) of the same object, holds paramount significance in the biological functionality of numerous molecules. This recognition has spurred increasing industrial interest in discriminating molecular chirality. A key challenge in this pursuit is augmenting the inherently weak circular dichroism (CD) chiroptical effect to facilitate efficient chiral sensing and separation. Over the past decade, the development of nanophotonic platforms designed for this purpose has garnered considerable attention [1].

Various strategies have been employed to address this challenge. One of them originates originating from the realms of plasmonics, exploiting the substantial field enhancement arising from the Surface Plasmon Resonance (SPR) phenomenon [2]. Another is the use of High Refractive Index Dielectric (HRID) materials due to their low-loss characteristics and magnetic resonances [3]. Hybrid structures, combining enhanced electric and magnetic resonances, have also been proposed, offering more advantageous chiroptical conditions for sensing applications [4].

In this context, we present our ongoing research on this subject. In particular, leveraging numerical simulations aided by multipole decomposition analysis, we illustrate how a straightforward hybrid gold-silicon metasurface can capitalize on the superposition of multiple resonances with non-radiating anapole states. This, results in remarkable conditions for enhancing the CD effect in the near-infrared spectrum. Furthermore, we demonstrate the direct relation of these anapole states with enhanced Third Harmonic Generation (THG), thereby opening avenues for background-free CD signals [5,6].

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